Chapter 7: QML Technique 2

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Table of Contents

1. Review
   - Quadratic unconstrained binary optimization (QUBO)
   - Ising model
   - Relationship between Ising models and QUBO
2. Adiabatic quantum computing
   - Adiabatic quantum computing
3. Quantum annealing
   - Quantum Annealing
   - D-wave

Table of contents

1. Review
   - Quadratic unconstrained binary optimization (QUBO)
   - Ising model
   - Relationship between Ising models and QUBO
2. Adiabatic quantum computing
   - Adiabatic quantum computing
3. Quantum annealing
   - Quantum Annealing
   - D-wave

Quadratic unconstrained binary optimization (QUBO)

- Quadratic unconstrained binary optimization (QUBO) is a combinatorial optimization problem with a wide range of applications from finance and economics to machine learning.[1]
- The QUBO problem is defined, for Q an $n \times n$ upper-triangular matrix of real weights and x a vector of n binary variables, as minimizing the function

$$f_Q(x) = \sum_{i=1}^{n} \sum_{j=1}^{i} Q_{ij} x_i x_j = \sum_{i=1}^{n} Q_{ii} x_i + \sum_{i<j} Q_{ij} x_i x_j$$

or, more concisely,

$$\min_{x \in \{0,1\}^n} x^T Q x.$$

[1] https://en.wikipedia.org/wiki/Quadratic_unconstrained_binary_optimization

Current subsection

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   - Ising model
   - Relationship between Ising models and QUBO
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   - Adiabatic quantum computing
3. Quantum annealing
   - Quantum Annealing
   - D-wave

Ising model

- The Ising Model is a mathematical model used for studying phase transitions and other properties of physical systems that evolve in time.[2]
- The problem concerns a collection of n particles arranged on the vertices of a graph $G = (V, E)$, which is often assumed to be a d-dimensional grid.
- Each particle can be in one of two states, called spins, represented by ± 1.

$$\mathcal{H}(\sigma) = \sum_{\langle i,j \rangle} J_{ij}\sigma_i\sigma_j - \mu \sum_j h_j\sigma_j$$

where real-valued parameters h_j, J_{ij}, μ for all i, j. J has the zero diagonal and symmetric. The spin variables σ_j are binary with values from $\{-1, +1\}$.

[2] McGeoch, C. C., *Adiabatic quantum computation and quantum annealing: Theory and practice*, Synthesis Lectures on Quantum Computing, 5(2), (2014) 1-93.

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   - Relationship between Ising models and QUBO
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   - Adiabatic quantum computing
3. Quantum annealing
   - Quantum Annealing
   - D-wave

- QUBOs are Ising models where the spin variables $\sigma_i \in \{-1, +1\}$ are transformed to binary-valued variables $x_i \in \{0, 1\}$.[3] This transformation $\sigma \mapsto 2x - 1$ yields an equivalent QUBO problem:

$$
\begin{aligned}
f(x) &= \sum_{\langle i,j \rangle} -J_{ij}(2x_i - 1)(2x_j - 1) + \sum_j \mu h_j (2x_j - 1) \\
&= \sum_{\langle i,j \rangle} (-4J_{ij}x_i x_j + 2J_{ij}x_j + 2J_{ij}x_j - J_{ij}) + \sum_j (2\mu h_j x_j - \mu h_j) \quad \text{using } x_j = x_j x_j \\
&= \sum_{i=1}^{n} \sum_{j=1}^{i} Q_{ij} x_i x_j + C
\end{aligned}
$$

where

$$
Q_{ij} = \begin{cases} -4J_{ij} & i \neq j \\ \sum_{\langle k\, i \rangle} 2J_{ki} + \sum_{\langle i\, l \rangle} 2J_{il} + 2\mu h_i & i = j \end{cases}, \quad C = -\sum_{\langle i\, j \rangle} J_{ij} - \sum_j \mu h_j .
$$

[3] https://en.wikipedia.org/wiki/Quadratic_unconstrained_binary_optimization

Table of contents

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   - Quadratic unconstrained binary optimization (QUBO)
   - Ising model
   - Relationship between Ising models and QUBO
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   - Adiabatic quantum computing
3. Quantum annealing
   - Quantum Annealing
   - D-wave

Introduction

- This section is based on the book *Adiabatic quantum computation and quantum annealing: Theory and practice* written by McGeoch, C. C..
- There are some computational models: Gate Models, Quantum Annealing, etc.
- Adiabatic quantum computation (AQC) was introduced by Farhi et al.[4].
- AQC algorithms are designed to solve optimization problems:
  Given an objective function $f : \mathbb{D}^n \to \mathbb{R}$ defined on n variables $x = x_1 \cdots x_n$ from some discrete domain $\mathbb{D}$, find an assignment of values to x that minimizes $f(x)$.

[4] [2] E. Farhi, J. Goldstone, S. Gutmann, J. Lapan, A. Lundgren, and D. Preda, *A quantum adiabatic evolution algorithm applied to random instances of an NP-Complete problem*. Science, vol. 292, no. 5516, 2001, pp. 472–475.

Algorithm description

- Setting: solve optimization problem P with objective function $f(x)$ works on a register Q of n qubits having superposition state $|\phi_t\rangle$ at time t.
- Components: a time-dependent Hamiltonian $\mathcal{H}(t)$
  1. An initial Hamiltonian $\mathcal{H}_I$ chosen so that the ground state of the system is easy to find.
  2. A final Hamiltonian $\mathcal{H}_F$ that encodes the objective function so that the ground state of Q is an eigenstate of $\mathcal{H}_F$ having minimum eigenvalue. A ground state s_g of Q corresponds to an optimal solution to P.
  3. An adiabatic evolution path, a function $s(t)$ that decreases from 1 to 0 as $t : 0 \to t_f$, for some elapsed time t_f. For now we use a simple linear path $s(t) = 1 - t/t_f$.
- $\mathcal{H}(t) = s(t)\mathcal{H}_I + (1 - s(t))\mathcal{H}_F$: an AQC algorithm for solving P.

Example

- 2-bit Disagree problem with the objective function $f(x_1x_2) = \begin{cases} 0, & x_1 \neq x_2 \\ 1, & o.w. \end{cases}$.
- The final Hamiltonian: $\mathcal{H} = \dfrac{1 + \sigma_1^z \sigma_2^z}{2}$.
- The initial Hamiltonian: $\mathcal{H}_I = \sum_{i=1}^{2} \dfrac{1 - \sigma_i^x}{2}$. whose the ground state is $|++\rangle$.
- AQC algorithm: $\mathcal{H}(t) = s(t)\mathcal{H}_I(t) + (1 - s(t))\mathcal{H}(t)$ where $s(t) = 1 - t/t_f$.
- Running: Set two qubits $Q = (q_1, q_2)$ having states expressed by $|\phi_t\rangle$ at time t and apply energies to the qubits according to $\mathcal{H}(t)$ for $t : 0 \to t_f$.
- End of computation: Q is in a low-energy superposition state $|\phi_T\rangle$ according to $\mathcal{H}$. Reading Q.
- According to the adiabatic theorem, if certain conditions hold, the algorithm will almost surely return an optimal solution.

Table of contents

1. Review
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   - Ising model
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   - Adiabatic quantum computing
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- [6][5] Quantum annealing (QA) is an optimization process for finding the global minimum of a given objective function over a given set of candidate solutions.
- Hamiltonian:

$$\mathcal{H}(t) = \mathcal{H}_F + \Gamma(t)\mathcal{H}_D$$

where $\mathcal{H}_F$ is the final Hamiltonian corresponding to the objective function and $\mathcal{H}_D$ is a transverse field Hamiltonian.
Here the transverse field coefficient Γ is initialized to some high value and gradually reduced to 0 over time.

[5] [6] https://en.wikipedia.org/wiki/Quantum_annealing

Current subsection

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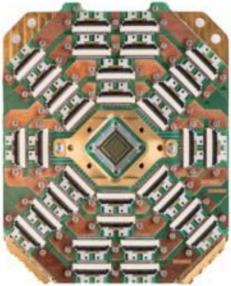
   - Adiabatic quantum computing
3. Quantum annealing
   - Quantum Annealing
   - D-wave

D-wave

- [5][6] D-Wave Systems Inc. is a Canadian quantum computing company. D-Wave was the world's first company to sell computers to exploit quantum effects in their operation.
  In 2019, D-Wave announced a 5000 qubit system available mid-2020.

(a) D-Wave Advantage System



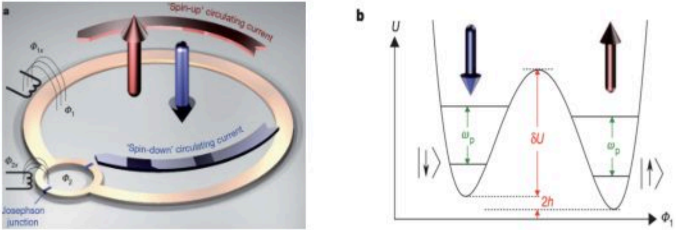
(b) Advantage Quantum Processing Unit

D-Wave[7]

[6] [5] https://en.wikipedia.org/wiki/D-Wave_Systems

[7] https://www.dwavesys.com/company/media-resources/

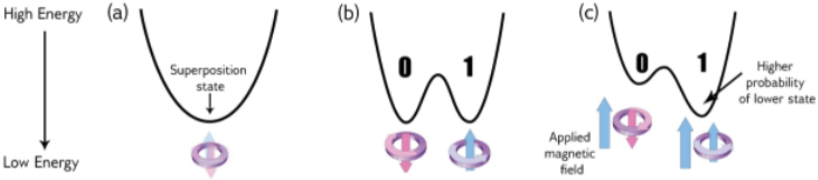
Quantum Annealing



Superconducting flux qubit. The figure is adopted from [4][8].

- [1] Each qubit is a loop of niobium controlled by Josephson junctions and biases represented by Φ's. The direction of current in the loop corresponds to the state of the qubit. Niobium loops can be place in superposition (current running in both directions) when supercooled to temperatures below 20 mK.

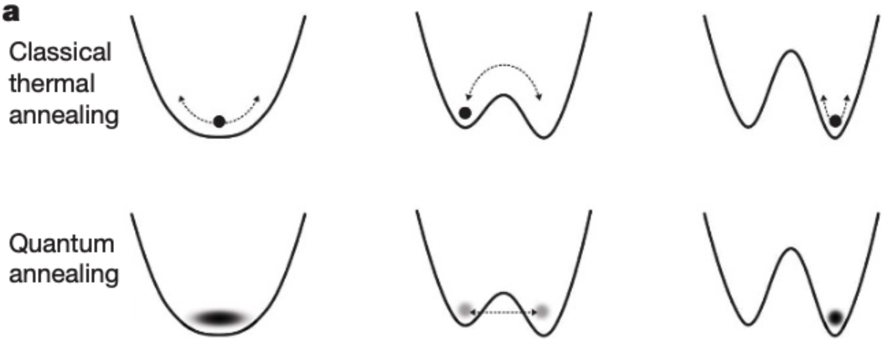
[8] [4] Johnson, M. W., Amin, M. H., Gildert, S., Lanting, T., Hamze, F., Dickson, N., ... and Rose, G.: *Quantum annealing with manufactured spins*. Nature, 473(7346), (2011) 194-198.



Energy diagram changes over time as the quantum annealing process runs and a bias is applied [8][9]

- one valley (a) goes to a double-well potential (b) when the quantum annealing process runs.
- One make (c) by applying an external magnetic field (called a bias).
- the qubit minimizes its energy in the presence of the bias.
- The programmable quantity:
  the external magnetic field, and the correlation weights between coupled qubits.

[9][8] https://docs.dwavesys.com/docs/latest/c_gs_2.html



Quantum annealing versus Thermal annealing [4][10]

[10][4] Johnson, M. W., Amin, M. H., Gildert, S., Lanting, T., Hamze, F., Dickson, N., ... and Rose, G.: *Quantum annealing with manufactured spins*. Nature, 473(7346), (2011) 194-198.

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[3] Schuld, M., and Petruccione, F., *Machine Learning with Quantum Computers*. 2021, Springer.

[4] Johnson, M. W., Amin, M. H., Gildert, S., Lanting, T., Hamze, F., Dickson, N., ... and Rose, G.:, *Quantum annealing with manufactured spins*. Nature, 473(7346), (2011) 194-198.

[5] https://en.wikipedia.org/wiki/D-Wave_Systems

[6] https://en.wikipedia.org/wiki/Quantum_annealing

[7] https://www.dwavesys.com/company/media-resources

[8] https://docs.dwavesys.com/docs/latest/c_gs_2.html

Thank you for listening.